

Flow of the CAN Protocol for Object Detection and Braking System

1. Object Detection Module

- Detects obstacles and sets the `Object_Detected` signal when an obstacle is within a threshold.
- Transmits the signal using the CAN protocol.

2. CAN Transmitter (TX) Module

- Encodes the `Object_Detected` signal into a CAN frame.
- Sends the CAN frame over the bus.

3. CAN Bus (Simulation in SystemVerilog)

- Transfers the CAN frame from the transmitter to the receiver.

4. CAN Receiver (RX) Module

- Decodes the received CAN frame.
- Extracts the `Object_Detected` signal.

5. Braking System Module

- Receives the `Object_Detected` signal.
- Activates braking if `Object_Detected` is asserted.

SystemVerilog Code for CAN Protocol and Object Detection Integration

1. Object Detection Module

```
module ObjectDetection (  
    input logic clk,  
    input logic reset,  
    input logic obstacle_in_range,  
    output logic Object_Detected  
);  
    always_ff @(posedge clk or posedge reset) begin  
        if (reset)  
            Object_Detected <= 0;  
        else  
            Object_Detected <= obstacle_in_range;  
        end  
    endmodule
```
